

Design, fabrication, and testing of an optical truss interferometer for the LISA telescope

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ABSTRACT

LISA is a space-based gravitational wave observatory aimed at detecting gravitational waves in the frequency range of 0.1 mHz to 0.1 Hz. The observatory is composed of three spacecraft, each separated by 2.5 million km in an equilateral triangle formation, trailing the Earth in a heliocentric orbit. One of the many crucial components to the mission is the LISA telescope, a bidirectional component used to expand an outgoing laser beam to the far spacecraft as well as compress a large incoming beam to a diameter of a few mm at the optical bench. Since the telescope is in the path of the long-baseline interferometer, its structure must be dimensionally stable at the $\text{pm}/\sqrt{\text{Hz}}$ level at mHz frequencies. A way to measure the stability of the LISA telescope is with a compact optical truss interferometer (OTI), consisting of three Fabry-Perot cavities mounted along the telescope to monitor structural distortions over time. All three cavities are operated with a common laser source, and each cavity is equipped with an acousto-optic modulator to shift the nominal laser frequency as well as an electro-optical modulator to modulate the laser phase for Pound-Drever-Hall locking. Variations in each cavity's length create variations in their corresponding laser frequency, which can be measured against a reference frequency that is locked to an external ultra-stable cavity. We will present the design and preliminary results in the fabrication and testing of first-generation OTI prototypes.

Keywords: LISA, OTI, optical truss, displacement interferometry, optical cavity, interferometer, optomechanical engineering, optical design

1. INTRODUCTION

1.1 LISA

The Laser Interferometer Space Antenna (LISA)^{1,7,9} is a mission led by ESA, in collaboration with NASA and an international consortium of scientists, to create a large-scale space-based gravitational wave observatory. The observatory will consist of a constellation of three spacecraft, each separated by 2.5 million kilometers to form an equilateral triangle in their formation. One of the main technological goals for LISA is to measure variations in the separation between each spacecraft with a sensitivity at the $10 \frac{\text{pm}}{\sqrt{\text{Hz}}}$ level between 0.1 mHz and 1 Hz using interferometric methods. The spacecraft will relay laser beams between each other which are used to measure the separation between freely falling test masses that are shielded, but untouched by the spacecraft. An incoming gravitational wave will induce perturbations in the separation between each pair of test masses, and these perturbations can then be measured and analyzed to determine the source and origin of the wave.

1.2 The optical truss interferometer

One of the most critical components to LISA are the six telescopes (two per spacecraft) that will be used to relay the laser beams. One proposed way to monitor the dimensional stability of the LISA telescopes is to use a Fabry-Perot based optical truss interferometer (OTI). The OTI system depicted in Figure 1 will be composed of three optical truss cavities (OTCs), each built in lengthwise at different positions around the telescope. Variations in the cavity lengths will in principle be solely due to variations in the telescope length along the three different lateral positions. We have developed a compact fiber-based Pound-Drever-Hall (PDH)² system that integrates a fiber-coupler, mode matching optics, and a cavity input mirror into a compact input stage for each OTC. These input stages will be attached around the primary mirror of the telescope, while the rear cavity mirrors (return stages) will be attached around the secondary mirror of the telescope. Using a PDH-like frequency locking technique, variations in the cavity lengths will change the frequency of the locked 1064 nm laser light. Thus, monitoring the frequency fluctuations in the laser light will in effect monitor displacements in

the telescope structure. The laser frequencies will be monitored using heterodyne detection by measuring a beat frequency between the laser light from each OTC and a laser that is locked to a stable reference cavity. The OTI project is a risk-mitigation plan to verify $\frac{\text{pm}}{\sqrt{\text{Hz}}}$ stability in the non-flight telescope units and, if necessary, can be used to monitor the telescope units that will fly with LISA in the case that pm-stability cannot be guaranteed.

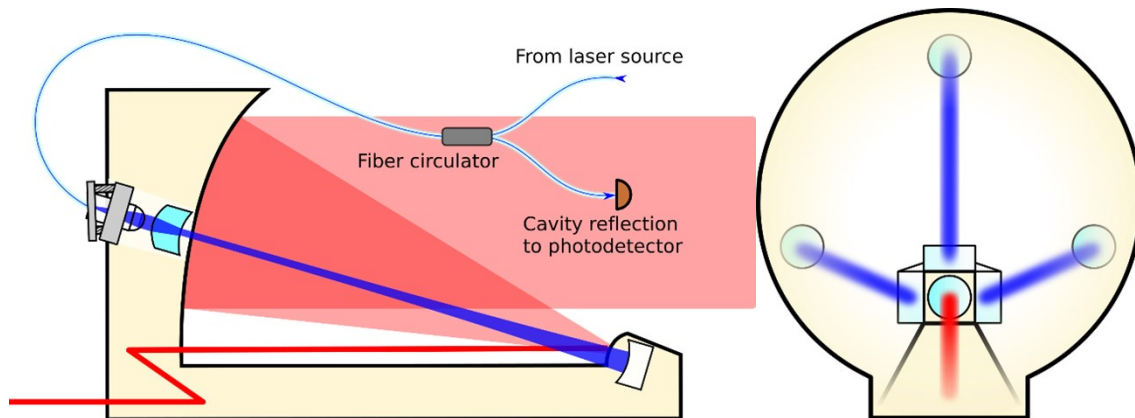


Figure 1. Conceptual schematic portraying how the truss cavities will be mounted onto the telescopes. On the left is a side view, where the OTC input stage can be seen attached to the primary (leftmost) mirror, and the return stage is attached to the secondary mirror. On the right is a head-on view, displaying how the three OTCs are to be mounted laterally around the telescope structure. Figure credit to Paul Fulda (UF).

In this paper, we will discuss our work on the design, fabrication, and testing of first-generation prototypes of the optical truss interferometer. We will go through our design methods in Section 2 and the fabrication and testing of prototypes in Section 3, followed by a discussion of current and future work on the project.

2. DESIGN METHODS

2.1 Initial cavity simulations

The choice of a cavity geometry for the OTC units is the first concern in the design process since many other aspects of the design are dependent upon both the cavity length and the curvature of the cavity mirrors. While the overall cavity length will ultimately be determined by the dimensions of the LISA telescope, a baseline length of 70 cm has been chosen as a starting point. The curvature of the cavity mirrors must allow for a stable cavity to occur, and the stability condition for this is shown below⁶.

$$0 \leq \left(1 - \frac{L}{R_1}\right) \left(1 - \frac{L}{R_2}\right) \leq 1 \quad (1)$$

Here L is the cavity length, while R_1 and R_2 are the radii of curvature of the first and second cavity mirrors, respectively. While there are many different types of cavities that satisfy the stability condition above, the two cavity geometries that we have investigated are the plane-concave and symmetric cavities. The most fitting cavity for the purposes of the project is that which is the most robust to misalignment of the cavity mirrors. The Frequency Domain Interferometer Simulation Software (FINESSE)^{3,4,5} has been utilized to simulate the response of various plane-concave and symmetric cavities to mirror misalignment. Early simulation results have shown that the symmetric cavity geometry has a lower sensitivity to mirror misalignments than the plane-concave geometry. Furthermore, Figure 2 below depicts the relationship between the cavity mirror curvature and the sensitivity of the system to mirror misalignments. The figure of merit that is used to determine the sensitivity of the overall cavity system is the visibility of the fringes in the reflected field. The optical power I that is reflected from the first cavity mirror has a dependence on the laser frequency that is incident upon the

cavity. The reflected power drops when the laser frequency is resonant with the cavity, and similarly the reflected power is a maximum when the frequency is anti-resonant with the cavity. . The cavity visibility is calculated by simulating a frequency sweep over the resonance of the fundamental mode and using the expression shown below.

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \quad (2)$$

This frequency sweep simulation is then iterated over a range of misalignment angles and cavity mirror curvatures to produce the plot below.

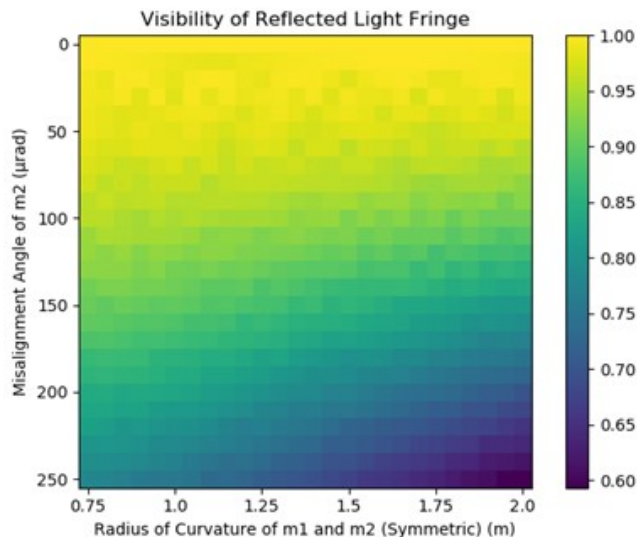


Figure 2. FINESSE simulation results showing the relationship between the mirror curvature in a symmetric cavity and the sensitivity of the visibility to misalignment of the mirrors. Here m_1 and m_2 refer to the first and second cavity mirrors, respectively. This analysis is only shown for misalignments of the second cavity mirror m_2 since this is a symmetric cavity, and the misalignment of m_1 would show the same results as seen above.

It can be seen from this analysis that a symmetric cavity with mirrors that have small curvature (large radii) will be more sensitive to misalignments than a symmetric cavity with mirrors that have large curvature (small radii). This also makes intuitive sense, since mirrors with very small curvature can mostly be regarded as flat, and a cavity with two flat mirrors will be very sensitive to their misalignment. On the other hand, there is an upper limit to the curvature that is allowed in a symmetric cavity which is determined by the cavity length. The maximum curvature that is possible in a stable symmetric cavity is when the radii of curvature of the mirrors are $R_1 = R_2 = \frac{L}{2}$, which is known as a concentric cavity⁶. Thus, the radius of curvature of the two cavity mirrors has been chosen to be 500 mm to allow for a maximum cavity length of 1 m.

2.2 OTC input stage

One of the main design goals for the OTI project is to develop a compact fiber-based input stage that integrates a fiber-coupler, mode matching optics, and the first cavity mirror into a single quasi-monolithic unit. This is a key aspect in the design due to the eventual need to bond these units onto the LISA telescope structure. The next step in the design process is then to develop a mode matching setup that will transform the transverse profile of an incident beam from an optical fiber into the fundamental cavity eigenmode. This includes a specific combination of a fiber, fiber-coupler, and mode matching lenses (Figure 3). Furthermore, the needs of the project demand that the length between the fiber-coupler and the first cavity mirror stays within a few cm so that these components can be assembled into a compact input stage.

We decided that the fundamental Gaussian mode output from the fiber-coupler will have a beam waist of 1 mm since the fiber components that would produce this mode are readily available to us. With this collimated output as a reference,

determining the necessary mode matching optics is a straightforward task. The mode matching calculations have been done using a combination of different software: JamMt⁸ and Zemax OpticStudio¹⁰. JamMt is a Java application that can be used to determine mode matching configurations based on the desired target mode, the incident mode, and a selection of lenses. Once a fitting mode matching configuration has been calculated, that optical system is then recreated in a Zemax environment. We then used the optimization function in Zemax to optimize the system to produce the desired cavity mode as well as to remain within a length of 5 cm. At this point in the design process, most of the optical design has been completed and what remains is to perform a tolerance analysis and to determine the feasibility of procuring the necessary optical components.

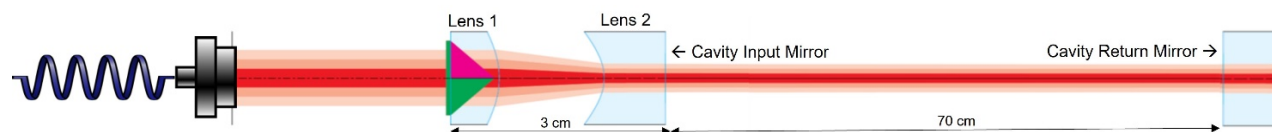


Figure 3. Schematic of the base OTC optical design. The length of the mode matching configuration is ≈ 3 cm excluding the size of the fiber-coupler, and the cavity input mirror is coated onto the back surface of the second mode matching lens. The reflectivity of both mirrors is at least 0.998 at 1064 nm, giving the cavity a finesse of ≈ 1600 . The integration of the second mode matching lens and the cavity input mirror allows for a more compact size. The OTC input stage refers only to the fiber-coupler, Lens 1, and Lens 2 in the schematic.

2.3 Tolerance analysis and compensation methods

We have done the tolerance analysis for the OTC design using both Zemax OpticStudio and FINESSE simulations. The tolerancing function in Zemax is utilized to perform Monte Carlo simulations that randomize the various misalignments and mismatches that will occur throughout the optical system due to imperfections in manufacturing and assembly. The data that is obtained from the Monte Carlo simulations is then exported from Zemax and written into a FINESSE environment to simulate the response of the cavity to these random variables. The results of the FINESSE simulations are then used to create a histogram as shown in Figure 4.

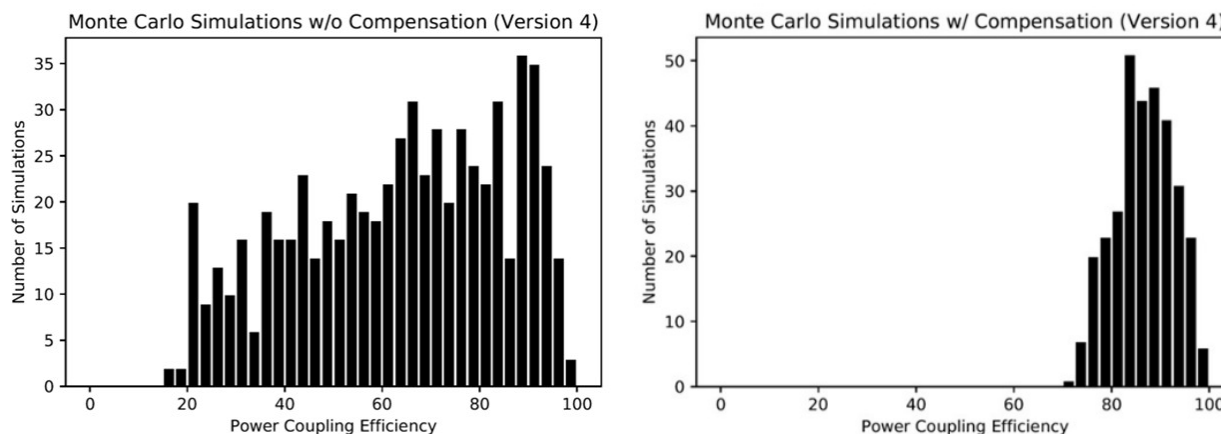


Figure 4. Histograms showing the results of Monte Carlo simulations of the optical truss cavity tolerances with and without compensation of the input beam. Compensation here refers to the adjustment of the collimated input beam, which will be discussed further below.

As can be seen on the left-hand side of Figure 4, the tolerances that have been used in these Monte Carlo simulations can have a severe impact on the performance of the system. To reduce the impact on performance, either the tolerances can be tightened, or compensation methods can be implemented if possible. Due to the precise nature of this optical system, the tolerances that have been set are already rather strict in comparison to manufacturing standards ($< 10 \mu\text{m}$ decenter, and

(< 1 arcmin tilt). Thus, the most effective way to reduce the impact on the performance of the optical system is to implement compensation methods. The primary compensation method that has been investigated for the OTC design is the adjustment of the fiber-coupler that is injected into the input stage. There are many commercially available fiber-couplers that have the degrees of freedom that are needed for the OTC. By adjusting the lateral position and tip/tilt of the input beam, misalignments throughout the input stage can be compensated to reduce their impact on the system. The compensation of the fiber-coupler can be modeled by writing a script into Zemax that interacts with the Monte Carlo simulations. Furthermore, the precision of the adjustments can also be written into the model to give a realistic picture of the compensation. The results of the Monte Carlo simulations with compensation of the input beam are shown on the right-hand side of Figure 4.

The secondary compensation method that has been investigated is the adjustment of the transverse mode that is output from the fiber-coupler. Along with the adjustment of the lateral position and tip/tilt, the fiber-coupler that we are using also has the capability of adjusting the distance between the fiber end-face and lens doublet, thus adjusting the transverse mode that is output from the fiber-coupler. While this adjustment will not compensate for misalignments in the input stage, it can be used to compensate for changes in the overall cavity length. A baseline cavity length of 70 cm has been set for the OTC, but this is not necessarily going to be the cavity length when the units are mounted onto the LISA telescope. An analysis has been made with FINESSE to show that the adjustment of the mode of the input beam can compensate for different overall cavity lengths, and the results of this analysis are shown below in Figure 5.

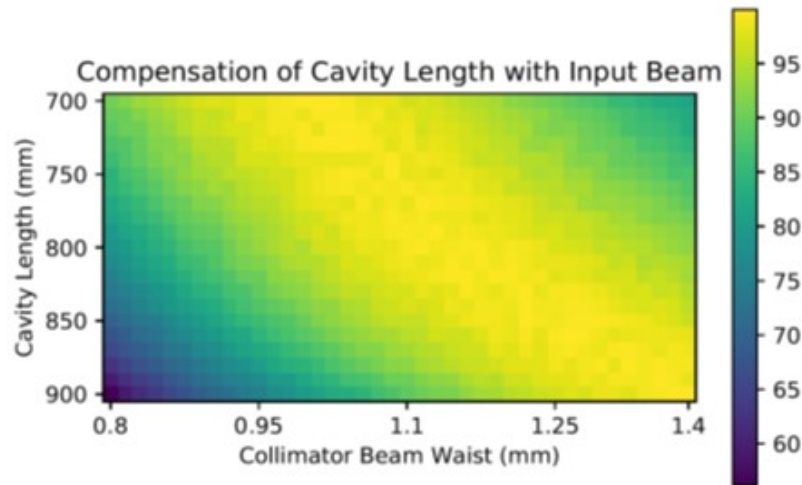


Figure 5. FINESSE simulation results showing that adjustment of the transverse mode of the input beam can compensate for different overall cavity lengths. This will allow us to assemble the OTC over a wide range of cavity lengths rather than being limited to only a 70 cm cavity.

3. FABRICATION AND TESTING

3.1 Fabrication Efforts

We have been working with a local manufacturer in Tucson, AZ to produce three first-generation OTC prototypes. Aside from the optical design, the mechanical design is critical to create a system that can measure displacements at the $\frac{\text{pm}}{\sqrt{\text{Hz}}}$ level. The materials that are used to house the optical components must have a very low coefficient of thermal expansion (CTE) such that thermal fluctuations in the environment produce as little displacement noise as possible. Such materials that are being used for fabrication of the prototypes are Schott's Zerodur glass-ceramic and Invar nickel-iron alloy, both of which have a CTE on the order of 10^{-6} K^{-1} or better. A graphical representation of the mechanical design of the input stage is shown below in Figure 6.

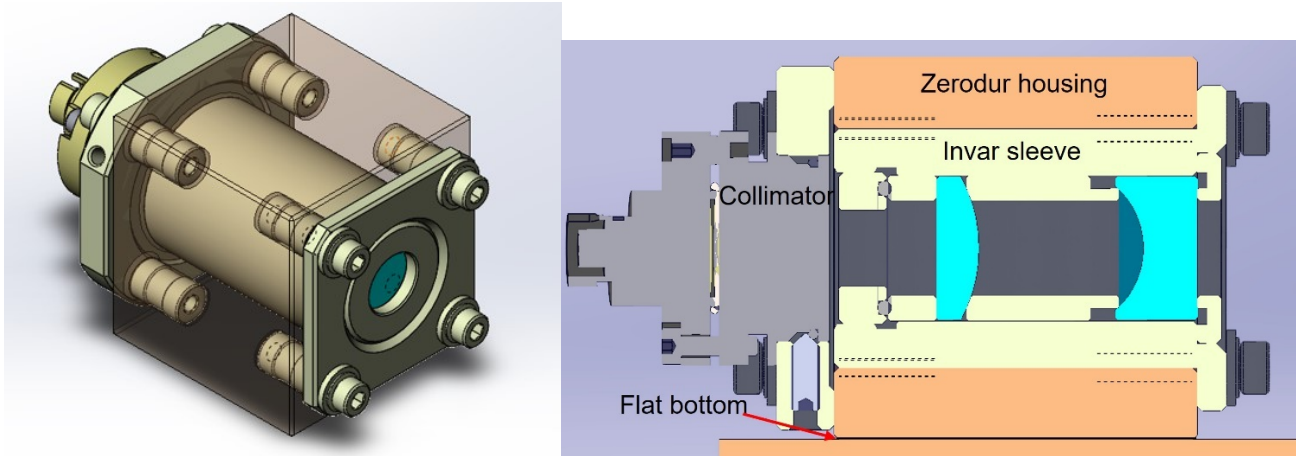


Figure 6. Graphical drawings of the input stage for the OTC prototype. An Invar sleeve, which holds the fused silica optical components in place, is inserted into a Zerodur housing which can then be bonded to a reference surface or telescope structure. The return stage has a similar design with the only difference being that the only optical component is a mirror.

The mechanical design of the input stage, as well as the return stage, is centered around creating a thermally stable housing for the optical components. There are spacer rings placed inside of the Invar sleeve that keep a proper separation between the lenses and make contact with their optical surfaces to center the optics with respect to each other during assembly. The retainer ring that is pressed against the cavity input mirror is designed such that the expansion and contraction of the Invar structure will have a minimal effect on the position of the vertex of the optical surface. This is a critical aspect of the design since the vertical distance between the optical axis and the flat bottom of the Zerodur housing must be stable to keep the cavity properly aligned. The flat bottom surfaces of both the input and return stages are polished to $\frac{\lambda}{10}$ ($\lambda = 633 \text{ nm}$) to allow for bonding of the units to either a reference surface or the LISA telescope structure. For purposes of testing the OTC prototypes, a long Zerodur baseplate is being procured to act as a flat reference surface to align the cavity on.

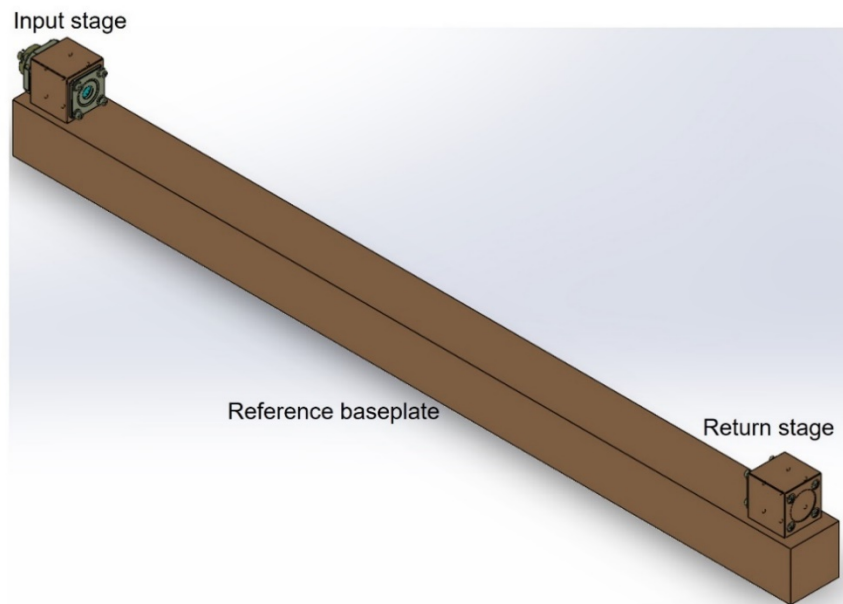


Figure 7. Graphical rendering of a prototype OTC mounted onto a reference baseplate. The baseplate being procured for testing will be wide enough to fit all three prototypes at once and must be flat enough to align the two OTC stages with respect to each other and observe a cavity visibility of at least 75%.

3.2 OTC prototype testing

Before the OTC prototypes are to be mounted onto a LISA telescope unit, they must be tested on a long, flat baseplate for a proof of concept. An important aspect of the prototype testing is to use a single laser source for all three cavities rather than using one laser per cavity. This can be done by utilizing a combination of an acousto-optic modulator (AOM) and an electro-optic modulator (EOM) that is accompanied with each OTC. The AOM is used to shift the nominal laser frequency and to tune that frequency shift to the resonance of the fundamental mode of each cavity. The EOM is used to modulate the phase of the beam for purposes of PDH frequency locking². However, in this type of PDH setup, the feedback signal from the control loop is to be fed back into the AOM rather than the original laser source. This will allow for three independent PDH locks to occur while stemming from the same laser source. The laser frequency from each cavity is monitored by heterodyne measurement of a beat frequency either using a stable reference frequency or by interfering the cavity beams with each other. The method of interfering the cavity beams with each other is only necessary in the absence of a stable reference frequency and requires a more in-depth analysis of the beat frequency measurements. Another important aspect of this system is that the beam that is reflected from the cavity must be able to couple back into the fiber so that the reflected signal can be measured from the output of an optical circulator. The schematic for this prototype testing system is depicted below in Figure 8.

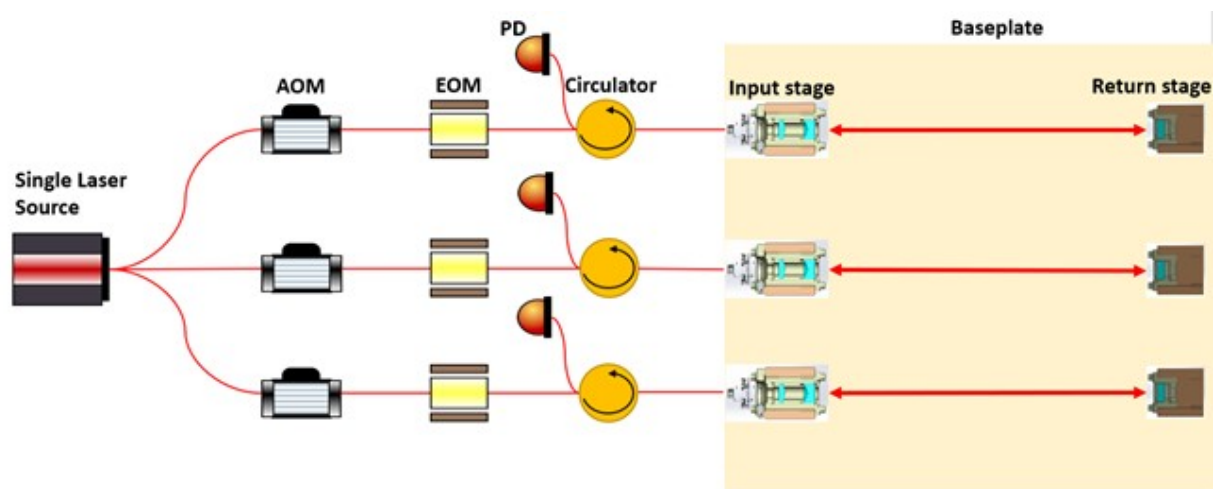


Figure 8. Schematic of a possible fiber based PDH system to test the three OTC prototypes. The reflected power from each cavity will travel back through the fiber circulator and be measured on the photodetectors to create the necessary signals for PDH frequency locking.

While the OTC prototypes have been in development, we have created a conceptually analogous benchtop system using commercially available components. This mock-OTC setup is being used to practice and demonstrate both the alignment methods as well as PDH locking techniques. A photo of this benchtop setup is shown below in Figure 9. The coupling of the reflected beam back into the fiber has been demonstrated with this system to be $> 85\%$, such that more than 85% of the power that is output from the fiber-coupler makes its way to the photodetector on the third port of the optical circulator. Furthermore, a cavity visibility of 40% has been achieved, and PDH locking techniques are currently being investigated.

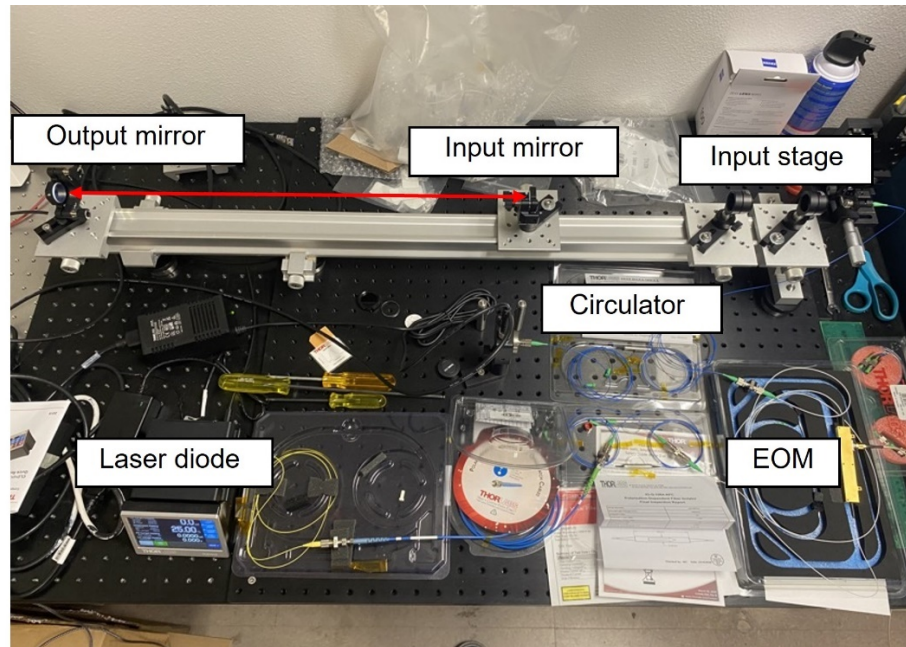


Figure 9. Overhead view of the mock-OTC benchtop setup which is similar to the fiber based PDH system that is depicted in Figure 8. A metallic optical rail is being used in place of the baseplate, and commercially available lenses and fiber components are used in place of an input stage.

4. CONCLUSION

The LISA telescopes are a critical component to the LISA mission as they lie directly in the optical path of the long-arm interferometer. The main purpose of the optical truss interferometer is to aid in the verification of pm stability in the telescope structure within the LISA frequency band. We have designed a 70 cm symmetric cavity with a finesse of 1600, and an input stage that integrates a fiber-coupler, mode matching lenses, and cavity input mirror into a single unit less than 5 cm in length. We have developed compensation methods that account for misalignments throughout the system as well as different overall cavity lengths. Furthermore, we are in the process of procuring mechanically and thermally stable optical housings that can be mounted onto stable, flat surfaces. The alignment methods and PDH locking techniques have been shown here and are planned for the testing of a first-generation OTI prototype later in 2021. The results of the prototype testing will verify that the OTI can measure displacements with pm resolution, and further work will be done to improve the system and prepare for testing of the LISA telescopes.

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REFERENCES

- [1] Amaro-Seoane, P., et al., “eLISA: Astrophysics and cosmology in the millihertz regime,” arXiv, 17 Jan 2012, <https://arxiv.org/abs/1201.3621>.
- [2] Black, E. D., “An introduction to Pound-Drever-Hall laser frequency stabilization,” American Journal of Physics 69, 79-87 (2001).
- [3] Brown, D. D., et al., “PYKAT: Python package for modelling precision optical interferometers,” SoftwareX 12, 100613 (2020).
- [4] Brown, D. D. and Freise, A., “Pykat,” Gwoptics, 1.2.1, July 2017, <http://www.gwoptics.org/pykat/>.
- [5] Freise, A., “FINESSE,” Gwoptics, 2.2, <http://www.gwoptics.org/>.
- [6] Siegman, A. E., [Lasers], University Science Books, 558-891 (1986).
- [7] Thorpe, J. I., “LISA Long-Arm Interferometry,” Classical and Quantum Gravity 27(8), 084008 (2010).
- [8] Thüring, A. and Lastzka, N., “JamMt,” ASR, 0.24, 25 Feb 2011, <http://www.sr.bham.ac.uk/dokuwiki/doku.php?id=geosim:jammt>.
- [9] Wanner, G., “Space-based gravitational wave detection and how LISA Pathfinder successfully paved the way,” Nature Physics 15(3), 200-202 (2019).
- [10] Zemax, “OpticStudio,” Zemax, 20.2.2, 2020, <http://www.zemax.com/>.